

CyberOps Command Center

Real-Time Cyber Threat Intelligence Platform



Project Vision & Objectives

The Challenge

Modern organisations face an escalating threat landscape with **over 2,200 cyberattacks occurring every day**. Traditional security monitoring tools lack real-time intelligence and centralised visibility, leaving SOC analysts overwhelmed and reactive rather than proactive.

The absence of integrated threat intelligence and automated analysis creates critical gaps in security operations, delaying response times and increasing organisational risk exposure.

Our Solution

CyberOps Command Center is a **web-based platform** that consolidates threat intelligence, asset monitoring, and network analytics into a single, unified interface powered by Firebase and AI integration.

Designed specifically for SOC analysts, the platform provides **real-time visibility** into security posture, automated threat detection, and intelligent insights that enable faster, more informed decision-making.



Core Objectives & Platform Features



Real-Time Threat Detection

Continuous monitoring and immediate alerting on suspicious activities across the entire network infrastructure



Centralised Dashboard

Unified view of all security metrics, threats, and assets through an intuitive, SOC-focused interface



AI-Powered Insights

Intelligent analysis using machine learning to identify patterns, predict threats, and recommend remediation actions

Key Features

01

Dashboard Overview

Real-time KPIs, threat radar, and traffic flow visualisation

02

Asset Intelligence

Comprehensive inventory with automated risk scoring

03

Threat Intelligence

MITRE ATT&CK mapping and threat feed analysis

04

Network Analytics

Traffic monitoring with flow analysis and anomaly detection

05

AI Agent

Natural language queries for instant security insights

Dashboard: Real-Time Security Operations

Visual Analytics

The centralised dashboard provides SOC analysts with immediate visibility into the security posture through **dynamic KPI cards** displaying asset counts, active threats, and risk scores.

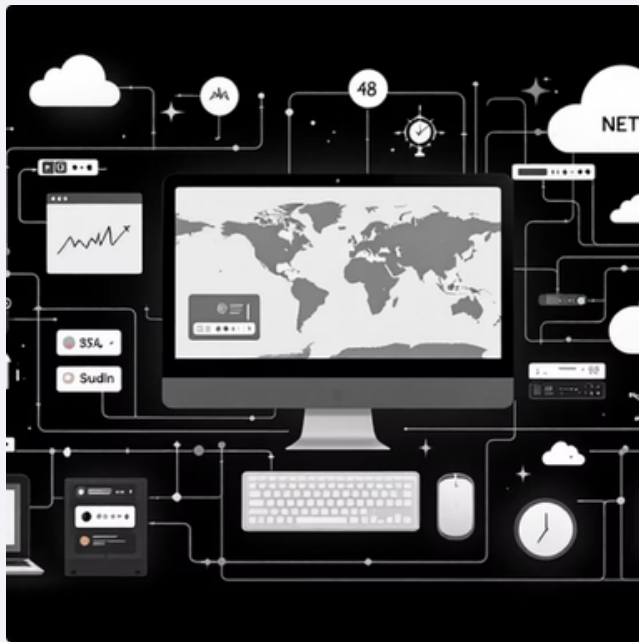
The threat radar visualisation maps attack vectors in real-time, whilst traffic flow trends reveal abnormal patterns through interactive pie, bar, and area charts.

Key Metrics

- Asset inventory with real-time status indicators
- Threat level scoring with colour-coded severity
- Network traffic analysis with anomaly detection
- Incident response time tracking



Intelligence & Analytics Modules



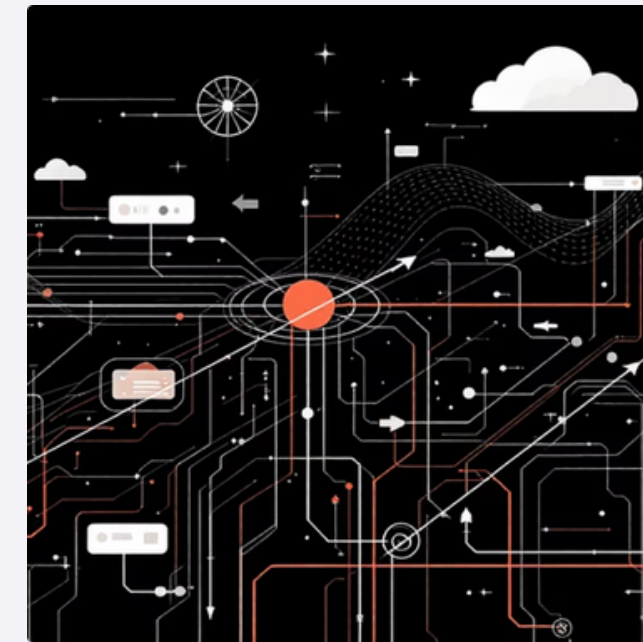
Asset Intelligence

Automated discovery and classification of all network assets with continuous vulnerability assessment. Each asset receives a **dynamic risk score** based on exposure, configuration, and historical threat patterns.



Threat Intelligence

Continuous ingestion and analysis of threat feeds from multiple sources. Integration with **MITRE ATT&CK framework** provides contextual mapping of observed tactics, techniques, and procedures.



Network Analytics

Deep packet inspection and flow analysis to identify malicious activity. **Behavioural baselines** detect deviations indicating potential compromise or lateral movement within the network.

Advanced Security Capabilities

Barcode & QR Scanner

Decode and analyse barcodes and QR codes encountered during security investigations to identify embedded malicious URLs or configuration data

Image Processing Suite

Extract metadata from images including **EXIF data** that may reveal location, device information, or hidden steganographic content concealing malicious payloads

Reconnaissance Tools

Built-in capabilities for network scanning, port discovery, and service enumeration to support ethical penetration testing and vulnerability assessment activities



AI Agent: Intelligent Security Assistant

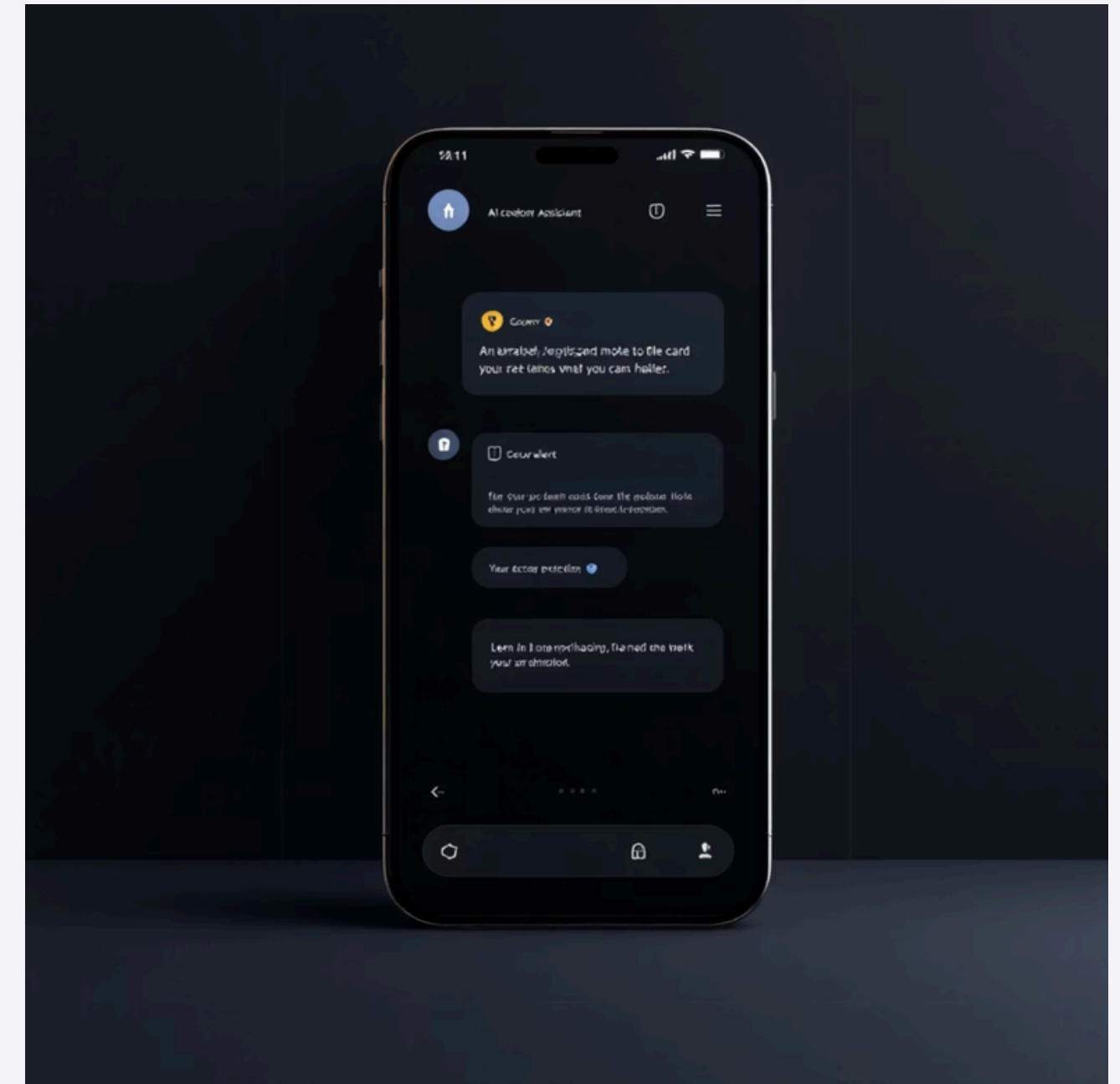
Natural Language Interface

The integrated AI agent transforms how SOC analysts interact with security data. Users can query the system using **plain English** rather than complex query languages or CLI commands.

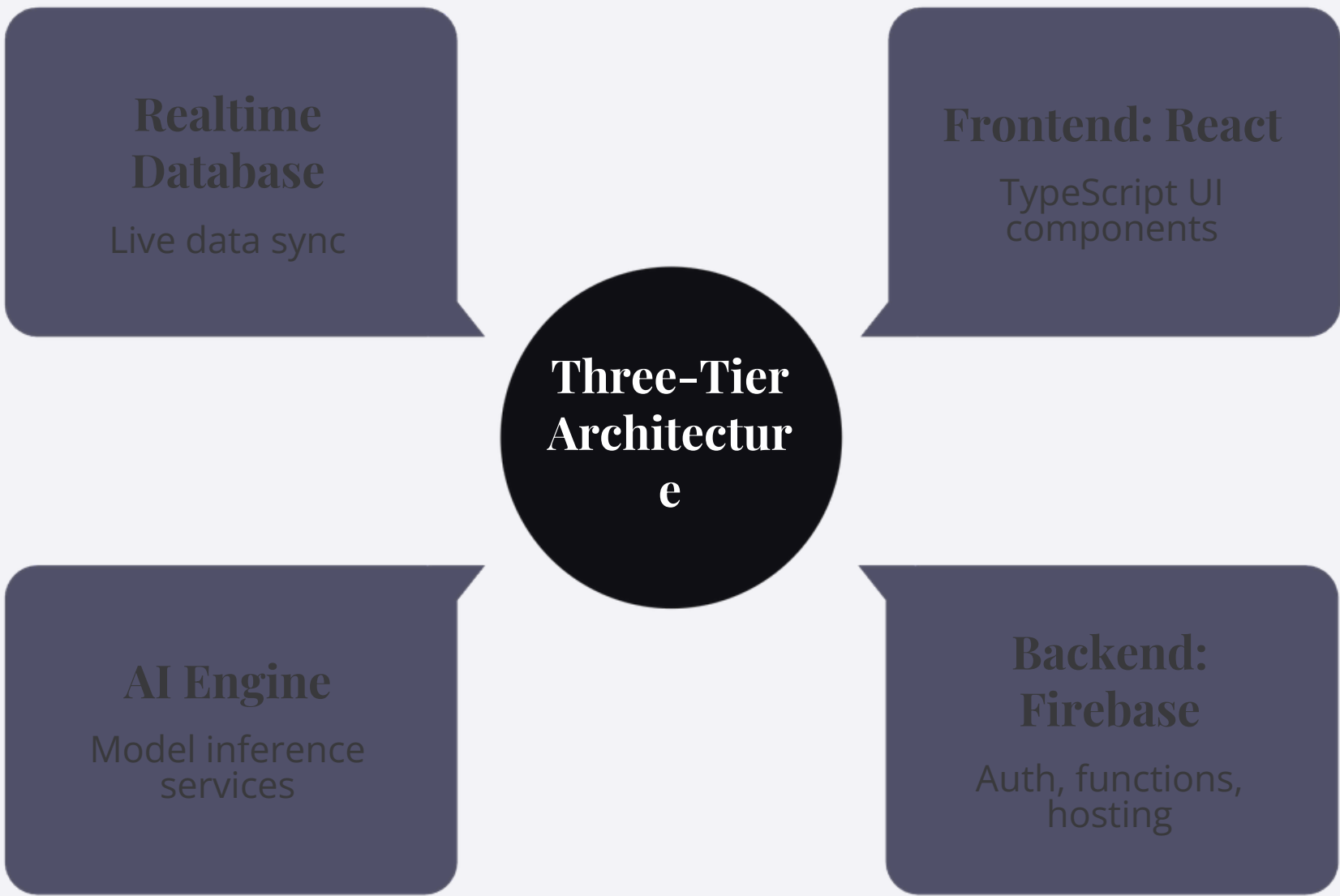
Example Queries

- "Show me all critical threats from the last 24 hours"
- "What assets are vulnerable to CVE-2024-1234?"
- "Generate incident report for IP 192.168.1.100"
- "Compare current traffic patterns with baseline"

The system responds with formatted answers, charts, and actionable recommendations, significantly **reducing mean time to respond (MTTR)** for security incidents.



System Architecture & Technologies



Technology Stack



Frontend Framework

React with TypeScript for component-based UI development and responsive design across all screen sizes



AI Integration

Custom machine learning models for threat prediction and natural language processing capabilities



Backend Platform

Firebase Authentication for secure user management and Firestore for real-time data synchronisation



Visualisation

D3.js and Chart.js for interactive data visualisations including radar charts, flow maps, and trend analysis

Platform Features & Future Enhancements

Current Capabilities

- Secure Authentication**
Role-based access control with encrypted credentials and session management
- Real-Time Status**
Live indicators showing system health, data freshness, and connection status
- Alert System**
Configurable notifications via email, SMS, or platform alerts with escalation protocols
- Error Handling**
Comprehensive logging with automatic recovery procedures and detailed audit trails

Future Roadmap

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Live Activity Feed
Streaming updates showing security events as they occur with user-configurable filters
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Enhanced RBAC
Granular permission system with multiple user roles and attribute-based access control
- 

Threat Prediction
Advanced machine learning models forecasting emerging threats based on global intelligence

Conclusion & Key Takeaways



Unified Platform

Consolidates threat intelligence, asset management, and network analytics into a single, intuitive interface



Real-Time Response

Reduces incident response time through automated detection and immediate alerting on security events



Data-Driven Security

Empowers SOC analysts with actionable insights derived from AI-powered analysis of security data

"CyberOps Command Center bridges the gap between overwhelming security data and actionable intelligence, enabling SOC teams to move from reactive firefighting to proactive threat hunting."

Impact on SOC Operations

The platform demonstrates **practical application of cybersecurity principles** including real-time monitoring, threat intelligence integration, and AI-assisted decision making. By reducing the complexity of security operations, organisations can **significantly improve their security posture** whilst optimising analyst productivity.

Future work will focus on expanding the AI capabilities, integrating additional threat intelligence sources, and implementing automated response workflows for common incident types.

Thank you for your time. For inquiries regarding this project, please contact the project team.